Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058,
India (Autonomous College Affiliated to University of Mumbai)

End Semester Examination
December 2025

Max. Marks: 100 **Duration: 3 Hrs.**
Class: T. Y. COMP, CSE **Semester: V**
Course Code: CE301 , CS301
Subject: Distributed Computing

Instruction:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary
(4) Mention the question number clearly while writing the answer
(5) Write answers to new questions on new page.

Course Outcomes: At the end of the course student will be able to:
1. **Understand the fundamental concepts and design principles of distributed systems.**
2. **Analyze and apply synchronization techniques and algorithms in distributed environments.**
3. **Evaluate consistency models and replication strategies for distributed data management.**
4. **Design and assess distributed file systems and understand their real-world applications.**
5. **Explore emerging trends and technologies in distributed computing.**

Q. No.		Max. Marks	CO BL
Q1 a	A smart city connects traffic lights, water systems, surveillance, healthcare, and public transport (nodes) using a shared data platform (middleware). Each subsystem operates semi-independently yet coordinates for global efficiency. a) Evaluate how well this scenario explains the interdependence and modularity of distributed systems. b) Critically assess the issues such as security, data consistency, and scalability in this scenario and how they parallel real distributed system issues. c) Defend whether a layered middleware architecture (application, platform, and network layers) is the best design choice for such an environment — justify with arguments.	10	1-5
Q1 b	A synchronous RPC call (single request and reply) has the following components: Client→Server request transmission = 4 ms. Server processing time = 10 ms. Server→Client reply transmission = 6 ms. Client marshalling overhead = 3 ms. Server unmarshalling + marshalling reply = 2 + 2 = 4 ms.	10	1-3

	Client unmarshalling = 2 ms. (a) Compute the total RPC response time (from application invoking RPC to getting result). (b) If the network has a 10% packet loss probability and lost requests are retransmitted after a timeout of 20 ms (timeout measured from original sending time), compute the expected RPC response time (assume only requests can be lost; if resend occurs, the same delays apply and server processes the request once when it receives it; ignore duplicate suppression).		
Q2 a	Assume you are designing a distributed banking application where multiple branches must exchange transaction records using message passing. Failures, delays, or message duplication can cause inconsistencies. a) Design a message-passing mechanism that guarantees *exactly-once delivery* and *ordering consistency* among branches. b) Propose how buffering, acknowledgment, and retransmission policies would operate to recover from partial failures. c) Create a conceptual diagram of your system and justify the design trade-offs (performance vs reliability).	10	1-6
Q2 b	Consider a hospital with mobile nurses (threads) attending patients (clients) across different wards (nodes). Each nurse carries patient records (code + state) and medical equipment (local resources). 1. Apply distributed thread concepts to model nurse movement across wards. 2. Compare strong vs weak code migration for moving nurses, highlighting the effect on medical equipment availability.	10	2-3
Q3 a	Consider 3 processes P1, P2, P3, initial vector clocks (0,0,0). The following events occur: 1. P1 does e1 (local) and sends m1 to P2. 2. P2 receives m1, does e2, sends m2 to P3. 3. P3 receives m2, does e3 and sends m3 to P1. 4. P1 receives m3 and does e4. Compute the vector clock for each event (show vectors) and state which events are concurrent and which are causally ordered.	10	2-3
Q3 b	A distributed system consists of N = 6 processes (P1–P6) implementing the Ricart–Agrawala algorithm for mutual exclusion. Each process requests access to the Critical Section (CS) once per round, and every request completes before the next begins. Network characteristics: Message transmission delay = 8 ms per message CS execution time = 50 ms Assume reliable communication and no overlapping requests. 1. Compute the total number of messages exchanged in the system for one complete round of all 6 processes accessing the CS once. 2. Calculate the average delay (in ms) experienced by a process before entering its CS, considering only communication delays. 3. Determine the total system time for one full round (i.e., all 6 processes finish one CS entry each).	10	2-3

Q4 a	Consider a chain of supermarkets that maintain local copies of the inventory database. Customers expect the availability information to be consistent across stores. 1. Map the supermarket scenario to data-centric consistency models: strict, sequential, causal, and eventual. 2. Analyze the trade-offs of strong consistency vs eventual consistency in terms of: Customer satisfaction, Network overhead and Update latency 3. Recommend a consistency model suitable for high-demand items and justify your choice.	10	3-4
Q4 b	A distributed online exam system has multiple servers. Some servers omit sending answers (omission failures) or delay sending responses (timing failures). 1. Apply the concept of omission and timing failures to this scenario. 2. Show how replication techniques can maintain system correctness and reliability. 3. Recommend a fault-tolerance mechanism to mitigate these failures.	10	3-3
Q5 a	You are designing a video streaming platform similar to Netflix that stores petabytes of video files using Hadoop Distributed File System (HDFS). 1. Apply your understanding of HDFS to describe how files are divided, replicated, and distributed across DataNodes. 2. Illustrate the role of NameNode and DataNode using a labeled diagram. 3. Analyze what happens when a DataNode fails and describe the recovery mechanism in HDFS.	10	4-3
Q5 b	A smart city uses sensors and cameras across intersections to manage traffic dynamically through Distributed AI and Machine Learning. Apply distributed ML concepts to show how edge devices and cloud servers collaborate for real-time decision-making. 1. Create a diagram showing data flow and model updates between sensors, local nodes, and central servers. 2. Describe how high-performance computing (HPC) can improve the learning and inference speed of this system. OR A smart campus has IoT devices for lighting, security cameras, and environmental sensors deployed across multiple buildings. The current cloud-only solution struggles with delays in controlling lights and alarms. (a) Apply the concept of fog computing to design a hierarchical architecture connecting IoT devices, fog nodes, and the central cloud. Describe the functions performed at each layer. (b) Illustrate how fog nodes can manage localized data processing and decision-making when network connectivity to the cloud is intermittent.	10	5-3